MK-312-BT v1.4 — Board Test Methodology

Systematic continuity & voltage checks to localize a build defect on a board that displays only solid blocks (no text). Tests verified against the v1.4 schematic and the BASCOM HC-05 firmware source which is known to drive this LCD correctly.

Setup before any test

1. **Power switch S5 OFF.**

2. **Disconnect the wall adapter from J8.** Battery may stay connected to J1B during voltage tests, but pull it for continuity tests so the board is fully de-energized.

3. **Pull the ATmega16 out of its socket** using a chip extractor or two flat-blade screwdrivers, alternating from each end. Set it on antistatic foam.

4. **Multimeter:** rotary to *continuity-beep* mode for tests 2/3/4, to *20 V DC* for test 1.

5. Identify a reliable **ground reference** on the mainboard: the **negative (–) battery terminal at J1B**. Every "GND probe" instruction in this document means "touch the J1B (–) post."

6. Test results in this document are derived from **schematic page 2 (AVR & J11)**, **schematic page 3 (front-panel J10/LCD/buttons)**, and the **BASCOM HC-05 init source** — the latter is ground truth because that firmware demonstrably drove this exact LCD model to readable text.

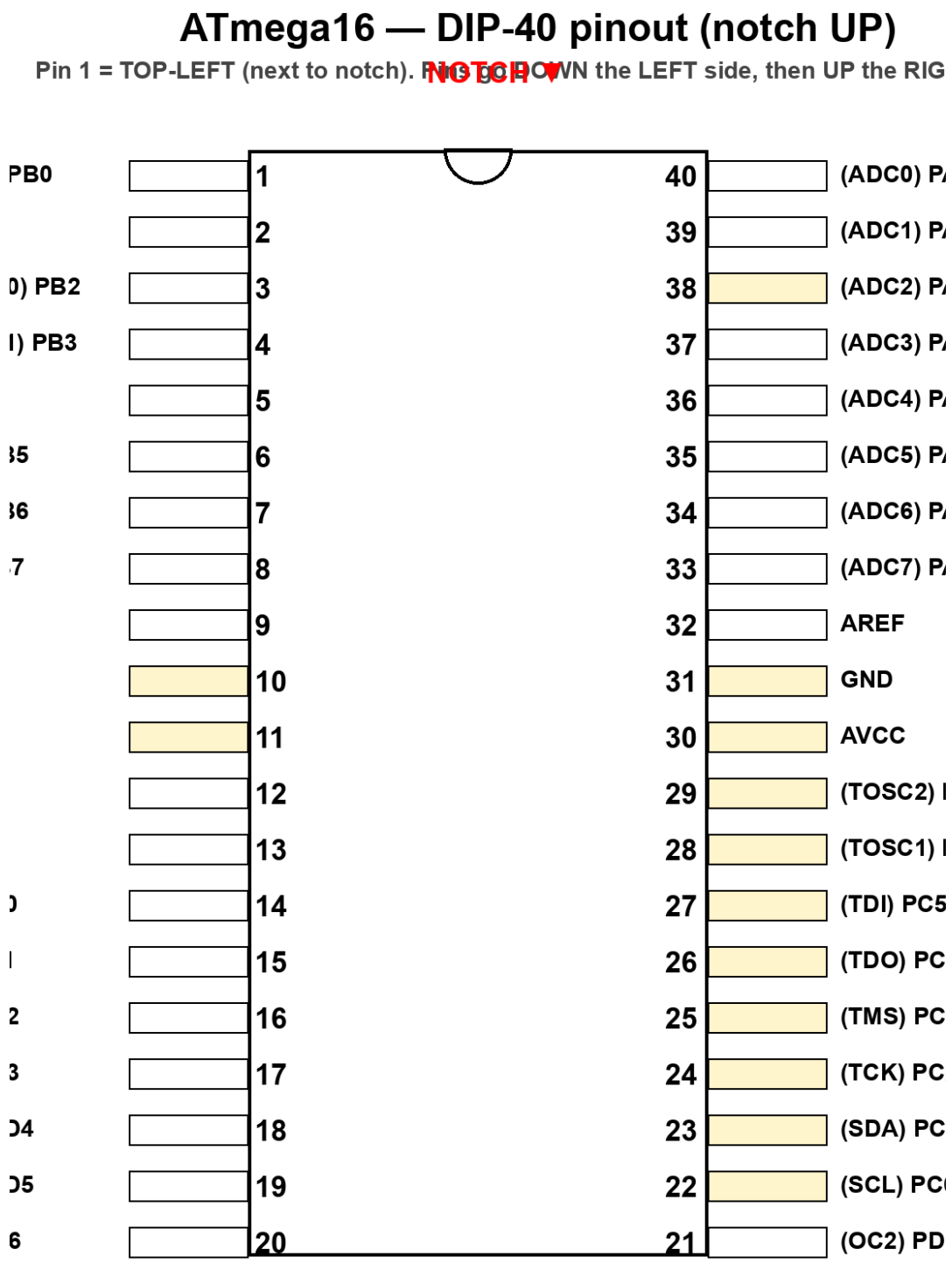
Symptom under investigation

Power on → LCD top row shows solid black blocks, bottom row blank. No text ever (one transient "Failure 21" once after a ribbon-flip event, never reproduced). User has confirmed: +12 V at JP4 = 12.9 V ✓, +5 V at JP3 = 5.0 V ✓, fuses LFUSE=0xFF / HFUSE=0xDC ✓, f005 firmware verified after flash ✓.

Therefore: power rails and the AVR are healthy. The fault has to be in the front-panel signal path, in LCD power/contrast, or in a solder bridge between PORTC pins on the mainboard. Tests below isolate which.

Reference 1 — ATmega16 pinout

The notch on the chip (and on the socket) marks the pin-1 end. With the notch oriented UP, pin 1 is top-left, pin counts go DOWN the left side, then UP the right side. Pins highlighted yellow are the ones referenced in the test procedures.



Reference 2 — LCD pinout (NHD-0216K1Z)

Newhaven NHD-0216K1Z, HD44780/ST7066U-compatible, 16-pin single inline header. Pin 1 is on the LEFT when looking at the LCD from the FRONT (text side). The back of the module usually has "1" and "16" silkscreened next to the end pins.

LCD module — NHD-0216K1Z (HD44780/ST7066U) — 16-pin SIL header

Pin 1 is on the LEFT when the LCD is viewed from the FRONT (text-readable side).
Pin numbers are usually printed on the back of the LCD next to pin 1 and pin 16.

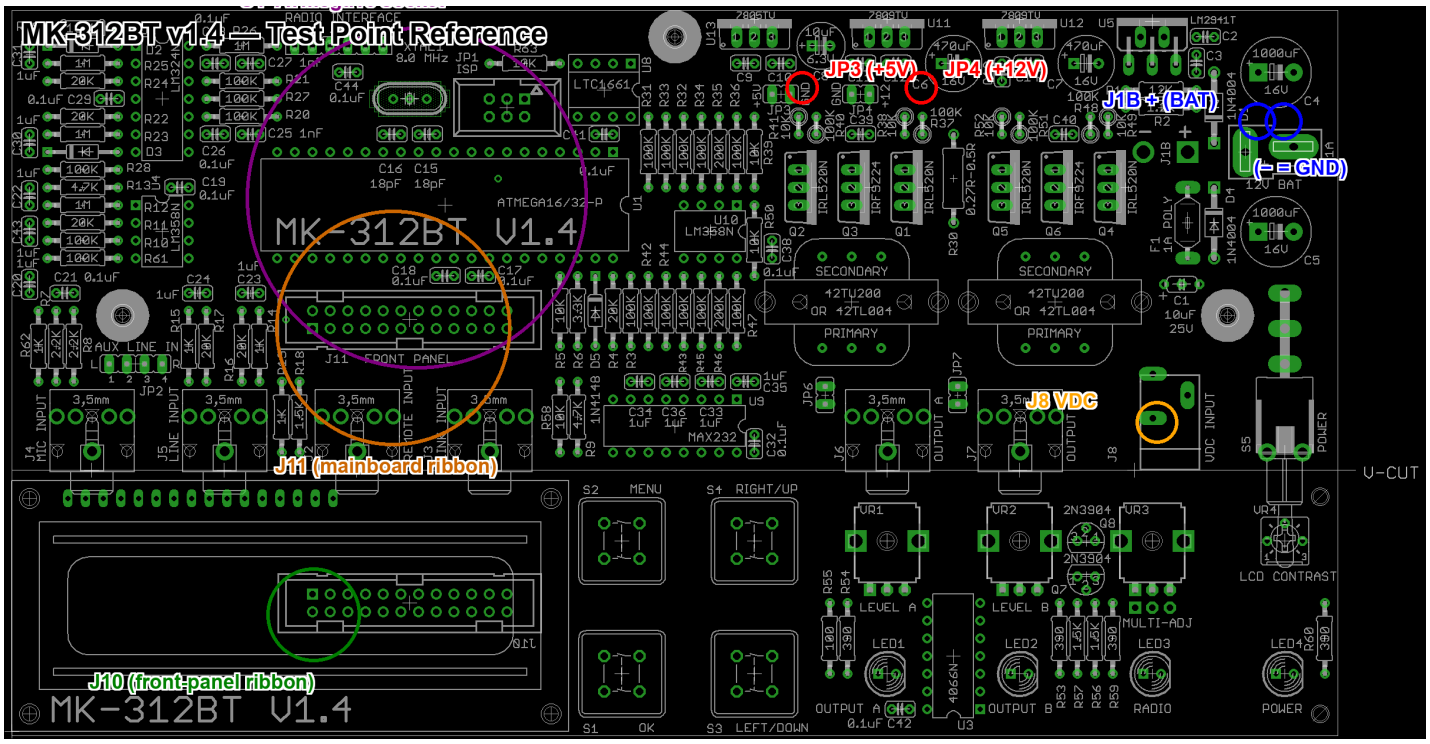
[LCD module — front side]

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

1	VSS	GND
2	VDD	+5V
3	V0	Contrast (VR4 wiper)
4	RS	from PC3 (AVR pin 25)
5	R/W	from PC1 (AVR pin 23)
6	E	from PC2 (AVR pin 24)
7	DB0	(unused, 4-bit mode)
8	DB1	(unused)
9	DB2	(unused)
10	DB3	(unused)
11	DB4	from PC4 (AVR pin 26)
12	DB5	from PC5 (AVR pin 27)
13	DB6	from PC6 (AVR pin 28)
14	DB7	from PC7 (AVR pin 29)
15	LED+	+5V via R55 (100Ω)
16	LED−	GND via Q8 transistor

Pin assignments above are taken from *schematic page 3* + the **BASCOM HC-05 firmware source**: `Db4=PC4, Db5=PC5, Db6=PC6, Db7=PC7, E=PC2, RS=PC3`. R/W is firmware-driven LOW on PC1 — it is **not** tied to GND on the board, so probing PC1 → GND should NOT beep (this is correct).

Reference 3 — Silkscreen overview (where the test points are)



Top half of the panel = mainboard. Bottom half = front panel (LCD + buttons + pots).

• **JP3 / JP4** — small 2-pin headers near the top, just below the linear regulators. JP3 = +5V/GND (the +5V rail test point). JP4 = +12V/GND (the +12V rail test point).

• **J1B (battery)** — two screw-terminal posts at top-right. Left = +12V battery, right = GND. The right post is the cleanest ground reference for everything in this document.

• **J8 (VDC INPUT)** — barrel jack on the right edge. Wall-adapter input. The center pin is +. Should read 0 V on battery-only operation.

• **U1 (ATmega16 socket)** — the 40-pin DIP socket in the middle of the mainboard, oriented with its long axis horizontal. The notch on the socket is on the LEFT.

• **J11** — 24-pin ribbon header on the mainboard, just below U1. Connects via ribbon to J10 on the front panel.

• **J10** — 24-pin ribbon header on the front-panel side. Other end of the ribbon. Same pin numbers as J11 (J11-1 ↔ J10-1).

Test 1 — Power rails (DC voltage)

Multimeter: 20 V DC. **Power switch S5: ON. AVR can stay in or out** for this test.

#	Black probe	Red probe	Expected	Pass / Fail
1.1	J1B (–) post	J1B (+) post	11.5 – 13.5 V	____
1.2	J1B (–) post	JP4 (+12V pad)	11.5 – 13.5 V	____
1.3	J1B (–) post	JP3 (+5V pad)	4.85 – 5.15 V	____
1.4	J1B (–) post	AVR socket pin 11	≈ 0 V (GND)	____
1.5	J1B (–) post	AVR socket pin 31	≈ 0 V (GND)	____
1.6	J1B (–) post	AVR socket pin 10	4.85 – 5.15 V (VCC)	____
1.7	J1B (–) post	AVR socket pin 30	4.85 – 5.15 V (AVCC)	____
1.8	J1B (–) post	LCD pin 1 (VSS)	≈ 0 V	____
1.9	J1B (–) post	LCD pin 2 (VDD)	4.85 – 5.15 V	____
1.10	J1B (–) post	LCD pin 3 (V0)	0.0 – 1.0 V (VR4 contrast)	____

If 1.10 reads > 1.0 V or stays at 5 V regardless of VR4 rotation: contrast pot wiper is shorted or the V0 trace from VR4 to LCD pin 3 is broken. With contrast at 5 V the LCD will show ALL solid blocks and never text — this is a leading suspect for the symptom.

If 1.6 or 1.7 ≠ +5 V: a trace from JP3 to U1 VCC/AVCC is broken. AVR cannot start.

Test 2 — AVR socket → LCD continuity (with ribbon connected)

POWER OFF, AVR REMOVED, RIBBON CABLE CONNECTED between J11 and J10. Multimeter to continuity-beep mode.

Each row tests one signal. Probe the SOCKET hole on the mainboard (where the AVR pin would land), and the LCD pin. The trace path is: AVR socket → mainboard trace → J11 pin → ribbon wire → J10 pin → front-panel trace → LCD pin. With a healthy build and correct ribbon orientation, this chain has < 5 Ω end-to-end and the meter beeps.

#	AVR socket pin	Signal	LCD pin	LCD signal	Should	Result
2.1	23	PC1 / R/W	5	R/W	BEEP	____
2.2	24	PC2 / E	6	E	BEEP	____
2.3	25	PC3 / RS	4	RS	BEEP	____
2.4	26	PC4 / DB4	11	DB4	BEEP	____
2.5	27	PC5 / DB5	12	DB5	BEEP	____
2.6	28	PC6 / DB6	13	DB6	BEEP	____
2.7	29	PC7 / DB7	14	DB7	BEEP	____
2.8	22	PC0 / mux	—	(to U3 4066 B-pins, not LCD)	no LCD pin	n/a

If ANY of 2.1 – 2.7 fails to beep: there is a break in that signal's path. Most likely causes, in order: (a) cold solder joint on a J11 or J10 pin (re-solder both connectors), (b) cracked PCB trace on the front panel, (c) ribbon cable wire broken (swap cable), (d) bent/missed pin in J11 or J10 receptacle.

Also worth checking — ribbon orientation: with both ends in the correct orientation, J11-1 connects to J10-1 (and J11-21 = +5V on both ends). If the ribbon is reversed, +5V on J10-21 would map to J11-4 (PC1) — which would push voltage backwards into the AVR's pin and is exactly the kind of event that could trigger a transient "Failure 21" only on the first switch-on after a flip. Verify red stripe on the cable is on the same end (pin 1 side) of both connectors.

Test 3 — Solder-bridge check between adjacent socket pins

POWER OFF, AVR REMOVED. Continuity-beep mode.

Probe two ADJACENT socket holes. Should NOT beep. A beep here means molten solder bridged between two AVR pins underneath the socket — common cause of "all blocks, no text" because the AVR ends up driving multiple LCD lines together.

#	Pin A	Pin B	Signals	Should beep?
3.1	22	23	PC0 (mux) — PC1 (R/W)	NO
3.2	23	24	PC1 (R/W) — PC2 (E)	NO
3.3	24	25	PC2 (E) — PC3 (RS)	NO
3.4	25	26	PC3 (RS) — PC4 (DB4)	NO
3.5	26	27	PC4 (DB4) — PC5 (DB5)	NO
3.6	27	28	PC5 (DB5) — PC6 (DB6)	NO
3.7	28	29	PC6 (DB6) — PC7 (DB7)	NO
3.8	29	30	PC7 (DB7) — AVCC	NO (would short DB7 to +5V)
3.9	30	31	AVCC — GND	NO (would dead-short the AVR)

If ANY pair beeps: visually inspect the underside of the AVR socket with a magnifying glass and good light. Look for a tiny solder ball or whisker between adjacent pads. Reflow with flux + braid to remove. Re-test before proceeding.

Test 4 — LCD power, contrast, and backlight

POWER ON, AVR can be in or out. If the symptom is "all blocks, no text" and Test 1 passed, this test isolates contrast/backlight as the cause.

#	Step	Procedure	Expected
4.1	Backlight on?	Look at the LCD with the lights dim. Is the green/blue backlight visible?	YES (glow visible)
4.2	Contrast sweep	With the box on and showing blocks, slowly rotate VR4 (LCD CONTRAST) all the way one direction, then the o	Blocks should fade to nothing, then come back
4.3	V0 voltage	Multimeter 20 V DC. Black probe = J1B (–). Red probe = LCD pin 3 (V0). Sweep VR4.	V0 should change, sweep VR4 between ≈ 0 V and ≈ 5 V a
4.4	DB lines static	AVR removed. Multimeter 20 V DC. Probe LCD pin 11/12/13/14 (DB4–DB7) vs GND.	All four lines should float ≈ 0 V (no pull-ups o

Why this matters: a stuck-high V0 line (5 V at LCD pin 3) makes the controller render every dot black regardless of what data the AVR sends — it's a textbook "all blocks, no text" symptom that's purely analog and has nothing to do with firmware. If 4.3 fails (V0 won't sweep), check VR4 wiper continuity and the 100 Ω R55 between V0 and the +5V backlight rail.

Decision tree

Run tests 1 → 2 → 3 → 4 in order. After each, branch on the result:

Test 1 (rails) FAILED any row.

Don't proceed. Fix the rail first — there's no point checking signal paths until VCC/AVCC are correct. Most likely: cold joint on regulator pin or 7805 input cap.

Test 2 (AVR → LCD continuity) FAILED any row.

That signal's trace is broken. Resolder both ends of the implicated J11/J10 pin first (cheapest fix). If still broken, inspect the trace with a magnifier — a hairline crack between J11 and the AVR socket is the most common build defect on this board.

Test 3 (solder bridge) FAILED any row.

Reflow the AVR socket pads with flux. A bridge between PC4–PC7 will look like "all blocks" because two LCD data lines get tied together — the controller sees garbage in both nibbles of every byte.

Test 4 (LCD power) FAILED.

If V0 won't sweep with VR4, the LCD itself can't render text — fix this before assuming anything is wrong with the firmware or the AVR.

All tests PASS but symptom persists.

At that point the LCD signal path is electrically clean and the AVR is healthy. Two remaining hypotheses, in order of likelihood:

(a) **LCD module is dead or has wrong controller revision.** Swap the LCD with a known-good 1602 (any HD44780/ST7066U/SPLC780 will do) and retry. ~$5 part, fastest test.

(b) **LCD reset timing** — the f005 firmware does an HD44780 init sequence assuming the LCD is in its reset state at power-on. The Newhaven ST7066U datasheet requires Vcc to rise from 0.1 V to 4.5 V within 0.1–10 ms for its internal POR to fire reliably. If the v1.4 power supply rises slower (typical with 1000 µF bulk caps + ferrite to AVCC), the LCD enters an undefined state and ignores the init sequence. Workaround: a watchdog reset pulse on the LCD or a firmware re-init. See *docs/troubleshooting/debug-notes/failure-21-analysis.md §7.1* for the disassembly evidence.